Vyas Balajee

Tempe, AZ | U.S. Citizen | vyasbalaje@gmail.com | linkedin.com/in/vyas-balajee-7aba9229a | github.com/vyasbalajee

Education

Arizona State University, Ira A. Fulton Schools of Engineering **Tempe, AZ**
B.S. Computer Science; GPA: 3.75 *Expected May 2027*

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Architecture, Compiler Design, Database Systems, Machine Learning

Experience

SkipCourse (ASU Capstone, industry sponsor) **Sep 2026 – Present**
Backend Developer, AI Learning-Assessment Platform *Remote*

- Built a Python data-validation pipeline (∼70 checks) that verified **222,340** curriculum-standard records across 52 jurisdictions and 214 frameworks before Neo4j import, replacing a registration-gated source with an openly licensed one
- Audited the live Neo4j database and its 768-dimension vector index, finding a schema mismatch that dropped topics and tools from student knowledge graphs
- Found access-control gaps in Firestore rules and backend APIs and wrote the fixes, feeding the team's FERPA audit
- Mapped ∼45 Cloud Run microservices across 3 repos (upload → Pub/Sub → grading worker → Neo4j vector search); set up secure VM access via IAP tunnels and OS Login
- Designed an "academic intensity" scoring method for the Gemini pre-processor and split its upgrade into 53 tasks across 5 engineers with interface contracts

Etalvis **May 2025 – Present**
Full-Stack Developer & Course Consultant *Remote*

- Sole developer of a production quiz platform (React, Node/Express, PostgreSQL on Railway) with CI/CD on every push, serving live users at test.etalvis.com
- Designed the quiz engine: randomized and topic-based modes, a 1–10 difficulty scale, timed sessions, per-question timing and mid-quiz resume
- Implemented JWT auth (bcrypt, email OTP, rate limiting) and GitHub-style role-based access control with role presets, per-user permission overrides and permission-gated API middleware
- Automated operations with GitHub Actions (orphaned-image cleanup), plus database backups, secret rotation and versioned releases
- Helped launch 4 embedded systems courses by reviewing curriculum, assignments and toolchains

Projects

Chess Engine | *C++* **Fall 2024 – Present**

- Implemented board representation, legal move generation and game-tree search in C++, with a command-line interface to play and test against the engine
- Designed a handcrafted evaluation using material and positional heuristics; currently replacing it with a neural network evaluation

Technical Skills

Languages: C, C++, Java, Python, JavaScript, SQL, Bash
Web & Backend: React, Node.js, Express, REST APIs, JWT auth, HTML/CSS
Databases: PostgreSQL, MySQL, Neo4j, Firestore
Cloud & Tools: Google Cloud (Cloud Run, Pub/Sub), AWS, Docker, Linux, Git, GitHub Actions, Railway, CMake
ML: Supervised learning, model training, NumPy, Gemini API